



Society of Petroleum Engineers

SPE-229468-MS

Transformative Synergy: Integrating Unconventional Resources with Site Facilities

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The integration of unconventional resources with existing infrastructure and processes is an essential step to leverage such new opportunities and sustain competitiveness in the growing energy landscape. This paper will attempt to address this integration by outlining a strategic approach for the ongoing COMPANY's unconventional operations with the Site field facilities. The aim is to drive productivity, maximize resource utilization, and enable seamless conventional and unconventional asset execution for sustaining long-term growth as part of COMPANY's strategic and transformation venture.

INTRODUCTION

The global energy sector is undergoing a profound transformation, driven by the urgent imperatives of sustainability, operational efficiency, and technological innovation. This shift is not merely a response to environmental pressures but a strategic recalibration of how energy companies manage resources, infrastructure, and production timelines. COMPANY, as a leading energy producer, has embraced this transformation through a groundbreaking initiative that integrates unconventional oil and gas resources into the existing infrastructure of Site—one of its most mature and strategically important fields.

This initiative marks a significant departure from traditional development models. Rather than constructing standalone facilities for unconventional production, COMPANY has opted for a synergistic approach that connects unconventional reservoirs directly to Site's established systems. This integration is not just a logistical convenience; it represents a strategic evolution that maximizes asset value, eliminates operational redundancies, and accelerates time-to-market for new production streams. The unconventional resources in question—UC 01, developed in collaboration with Petronas, and UC 02, led by COMPANY—are situated near Site, making them ideal candidates for this integrated development strategy.

Unconventional reservoirs differ fundamentally from conventional ones in terms of geological characteristics, fluid dynamics, and extraction methodologies. Typically characterized by low porosity and permeability, these reservoirs do not allow hydrocarbons to flow freely into the wellbore. As a result, they require advanced stimulation techniques such as hydraulic fracturing, horizontal drilling, and enhanced reservoir contact to unlock their production potential. These methods are not only technically

demanding but also require precise engineering and operational coordination to ensure economic viability and environmental safety.

COMPANY's unconventional oil and gas assets are estimated to contain approximately 220 billion barrels of oil and 460 trillion cubic feet of gas. These figures underscore the strategic importance of developing unconventional resources as part of the company's long-term energy portfolio. To tap into these vast reserves, COMPANY is deploying cutting-edge technologies that have been refined through global best practices and localized expertise. Horizontal drilling allows for greater reservoir contact, while hydraulic fracturing enhances permeability and flow rates. Together, these techniques form the backbone of COMPANY's unconventional development strategy.

The economic rationale behind the rise of unconventional oil production is multifaceted. During periods of low oil prices, energy companies are compelled to innovate and improve extraction efficiency to maintain profitability. This often leads to the adoption of unconventional methods that can yield higher returns despite challenging reservoir conditions. Conversely, when oil prices remain elevated over extended periods, previously uneconomical reserves become viable targets. This shift in economic feasibility has catalyzed the development of advanced recovery techniques, including steam, gas, and chemical injection processes that fracture rock formations and release trapped hydrocarbons.

Recognizing these dynamics, COMPANY has launched two pivotal unconventional oil projects: UC 01 and UC 02. UC 01, developed in partnership with Petronas, benefits from international expertise and collaborative innovation. UC 02, managed by COMPANY Onshore, leverages local knowledge and operational agility. Both projects are strategically located near Site, enabling seamless integration into existing infrastructure. This proximity allows COMPANY to avoid the capital-intensive process of building new facilities, instead utilizing Site's robust systems for processing, transportation, and storage.

The integration of UC 01 and UC 02 into Site is supported by a comprehensive operational and organizational framework. This framework encompasses reservoir engineering, production logistics, surface facility optimization, and stakeholder coordination. It is designed to address the unique challenges posed by unconventional reservoirs while ensuring alignment with COMPANY's broader strategic objectives. Key components of this framework include real-time data monitoring, predictive maintenance systems, and adaptive production planning—all of which contribute to enhanced efficiency and reduced downtime.

From a technical perspective, the integration process involves several critical steps. First, reservoir characterization is conducted using advanced seismic imaging and petrophysical analysis to determine the most effective drilling and stimulation strategies. Next, well design is optimized to maximize reservoir contact and minimize environmental impact. Surface facilities are then adapted to handle the specific flow characteristics of unconventional hydrocarbons, which often include higher levels of gas, water, and particulates. Finally, production data is continuously analyzed to refine operational parameters and improve recovery rates.

Operationally, the integration requires close collaboration between multiple teams, including drilling, completions, production, and facilities engineering. Cross-functional coordination ensures that each phase of the development process is aligned with project goals and timelines. COMPANY has also implemented rigorous safety and environmental protocols to mitigate risks associated with unconventional production. These include well integrity monitoring, emissions control systems, and water management strategies that adhere to international standards.

Organizationally, the success of this initiative depends on effective leadership, clear communication, and a culture of innovation. COMPANY has established dedicated project teams for UC 01 and UC 02, each tasked with overseeing specific aspects of the integration. Regular review meetings, performance dashboards, and feedback loops enable continuous improvement and agile decision-making. The involvement of strategic partners like Petronas further enhances the project's capabilities by bringing in global perspectives and technical expertise.

The insights gained from integrating UC 01 and UC 02 into Site offer valuable lessons for future unconventional projects within COMPANY's portfolio. These include the importance of leveraging existing infrastructure, the need for advanced reservoir characterization, and the benefits of cross-functional collaboration. By documenting these methodologies and outcomes, COMPANY is creating a replicable model that can be applied to other fields and resource types.

In conclusion, COMPANY's initiative to merge unconventional resources with Site's conventional infrastructure exemplifies a forward-thinking approach to energy development. It reflects a deep understanding of geological, technical, and economic factors, as well as a commitment to sustainability and operational excellence. As the energy sector continues to evolve, such integrated strategies will play a crucial role in shaping the future of oil and gas production—ensuring that resources are developed responsibly, efficiently, and profitably.

METHODOLOGIES

The integration of Unconventional Projects UC-01 (Petronas) and UC-02 (COMPANY) into the Site facilities marks a strategic milestone in optimizing upstream production capabilities. This initiative begins with a clear demarcation of project boundaries, ensuring that all operational, technical, and strategic elements are aligned. The scope encompasses the incorporation of initial production facilities, multiphase pumping systems, pipeline tie-ins, and robust measurement strategies for both oil and gas flows. These components are critical to maintaining flow assurance and operational integrity across the integrated network.

The integration process is initiated through a comprehensive feasibility study that evaluates the compatibility of UC production blocks with the existing Site infrastructure. This study assesses technical constraints, capacity limits, and safety protocols, laying the groundwork for a seamless transition. Once feasibility is confirmed, the project advances to the engineering design phase, where interface points are established and a RACI (Responsible, Accountable, Consulted, Informed) matrix is developed. This matrix ensures clarity in roles and responsibilities, enabling efficient coordination across multidisciplinary teams.

Production testing on the drilled UC-02 wells has provided valuable operational insights. These tests validate flow characteristics, reservoir behavior, and equipment performance, forming the basis for rigorous operational protocols. The data gathered supports the implementation of best practices in flow control, pressure management, and production optimization. These protocols are designed to ensure system reliability and maximize hydrocarbon recovery while minimizing environmental impact.

A key success factor in this integration is the collaboration among diverse teams—unconventional operations, process engineering, and Site field operations. This multidisciplinary approach fosters innovation and agility, allowing the project to adapt to evolving challenges. Regular cross-functional reviews and technical workshops facilitate knowledge sharing and alignment on project goals.

Insights from early production experiences at UC-02 wells, which have already been reintroduced into Site facilities, serve as a valuable reference for future integration efforts. These experiences highlight the importance of proactive planning, real-time monitoring, and adaptive control strategies. Lessons learned include the need for flexible pipeline routing, enhanced metering accuracy, and contingency planning for flow disruptions.

This paper presents a multi-stage, field-integrated methodology for connecting unconventional oil developments to legacy infrastructure. The approach is structured around key pillars: boundary definition, feasibility analysis, engineering design, operational testing, and collaborative execution. Each stage is supported by data-driven decision-making and continuous improvement practices.

By documenting the challenges faced and solutions implemented during the integration phase, this framework offers a replicable model for future unconventional resource developments. It emphasizes

strategic alignment, technical rigor, and operational excellence—principles that are essential for maximizing value generation and ensuring long-term sustainability in complex field environments.

Feasibility Study

A comprehensive feasibility study was conducted to assess the integration of unconventional resources (UC) into the existing Site infrastructure. The evaluation focused on three critical dimensions: infrastructure compatibility, production capabilities, and operational risk management. The findings revealed that the current Site facilities are well-equipped to accommodate UC production with minimal to no structural modifications. This indicates a high level of readiness and adaptability within the existing setup, which significantly reduces the need for capital-intensive upgrades or redesigns.

However, the study emphasized that interface points—where UC streams connect with existing systems—require meticulous engineering attention. These junctions are vital to maintaining flow assurance and ensuring operational safety. Engineering teams must address potential challenges such as pressure differentials, fluid compatibility, and control system integration to prevent bottlenecks or safety hazards. The analysis also highlighted the importance of implementing robust monitoring and control mechanisms at these interface points to detect anomalies early and maintain system integrity.

Overall, the feasibility analysis supports the strategic integration of UC resources into Site operations, reinforcing the facility's capability to enhance production timelines and sustainability goals while maintaining safety and efficiency standards. Let me know if you'd like this formatted for inclusion in this paper or presentation.

Reservoir Appraisal and Pilot Production

Reservoir potential was assessed through three production phases (UC 02):-

- **Phase-1A:** - Initial production from Pad HUP-02, incorporating flowback testing and directing the output to the suction side of SITE Facilities MPP.
- **Phase-1B:** - Construction of 11 wells at FLP-01, with the output routed to the discharge of SITE Facilities MPP, which includes metering for both oil and gas.
- **Phase-II & III:** - Future tie-in to one of SITE plats planned for subsequent expansion.

Facility Design & Engineering

The design phase of the facility was marked by a comprehensive and methodical approach aimed at ensuring seamless integration of unconventional production wells into the existing Site infrastructure. This phase began with meticulous planning and execution of pipeline route surveys, which laid the groundwork for the subsequent detailed engineering activities. These engineering efforts focused on establishing robust connections between the new wells and the Site's surface amenities, ensuring optimal flow and operational efficiency.

A critical component of this phase was the development of Hub facilities, strategically positioned to manage production handling and facilitate the smooth transfer of hydrocarbons. These hubs were designed not only to support current production volumes but also to accommodate future scalability, reflecting a forward-looking operational strategy.

Integration efforts were particularly concentrated on harmonizing essential systems such as power supply, lighting, and signal infrastructure. This alignment with existing Site operations was vital to maintain continuity and avoid disruptions during commissioning and startup. Each system was evaluated for compatibility and resilience, ensuring that the new installations complemented the legacy infrastructure.

Throughout the design phase, safety remained the highest priority. Detailed safety assessments were conducted for every asset involved, with a strong emphasis on risk mitigation and compliance with Company's operational standards. Execution activities were planned with precision to uphold safety

protocols, minimize hazards, and ensure secure working conditions for all personnel. This rigorous safety-first approach underscored the project's commitment to operational excellence and sustainable development.

Production Testing

The production testing phase for the UC 02 wells represented a significant milestone in verifying the viability of incorporating unconventional resources into Site's conventional infrastructure. The testing period for the ready wells was specifically started to evaluate flow performance, reservoir behavior, fluid compatibility, and system response under actual operational conditions.

Preliminary results indicated that the early-stage wells of UC 02 achieved stable flow rates, and the key operational insights gained during the testing included:-

- **Equipment performance:** - Surface facilities such as flowlines and pumps functioned within design specifications, confirming the engineering assumptions established during integration planning.

Operational readiness was attained through comprehensive testing of the surface pumps to confirm their capacity and ensure redundancy. In partnership with the crisis management team, Emergency Response Procedures were formulated to mitigate potential risks.

In summary, the production testing of UC 02 validated the technical feasibility of integration efforts while laying a robust foundation for full-scale operations. It further emphasized the critical role of early-stage testing in mitigating risks associated with unconventional developments and ensuring smooth integration with existing infrastructure.

Performance Monitoring Data Collection

Production data was systematically collected across multiple operational phases, capturing cumulative oil volumes and detailed flowback periods. To ensure precise tracking of both liquid and oil flows, a combination of fiscal metering systems and ultrasonic flowmeters was employed. These technologies provided high-resolution measurements essential for evaluating reservoir behavior and optimizing production strategies. The data enabled engineers to monitor real-time performance, identify anomalies, and adjust operational parameters to enhance recovery efficiency.

Additionally, the integration of these metering tools supported compliance with regulatory standards and internal operational benchmarks. By analyzing flow trends and cumulative outputs, the team could refine reservoir models, forecast production potential, and align field activities with strategic objectives. This approach not only improved decision-making but also reinforced the reliability of production reporting and asset performance evaluations.

Multidisciplinary Collaboration

A multidisciplinary team comprising unconventional operations, process engineering, and site operations collaborated closely to ensure the seamless implementation of the initiative. This cross-functional synergy was instrumental in aligning strategic objectives with operational execution. Regular alignment meetings served as a vital platform for knowledge exchange, enabling the team to proactively address technical and logistical challenges. Coordination was maintained through structured mechanisms such as RACI matrices, which clearly defined roles and responsibilities, and continuous feedback loops with concession holders to ensure transparency and responsiveness. These feedback loops facilitated real-time decision-making and helped mitigate risks by incorporating stakeholder insights at every stage. The active engagement of all stakeholders—ranging from field engineers to concession partners—was essential in maintaining alignment, accelerating approvals, and fostering a shared sense of ownership. This collaborative framework not only enhanced operational efficiency but also ensured that implementation timelines were met without

compromising safety or quality. The initiative stands as a testament to the power of integrated teamwork and disciplined project governance in driving successful outcomes in complex operational environments.

CHALLENGES AND MITIGATION STRATEGIES

During the integration of UC 01 and UC 02 into the Site facilities, a range of technical and operational obstacles arose. Addressing these issues necessitated vigilant planning, collaboration across various functions, and customized mitigation strategies to achieve effective implementation. Table 1 outlines the principal challenges faced along with the respective measures taken to address them throughout the project's duration.

Table 1—Challenges and Mitigations

Challenge	Mitigation
Infrastructure compatibility	Feasibility and interface study
Cross company coordination	RACI, AUA, and SLA.
Safety and compliance	Integrated HSE Matrix
Fluid compatibility	Wells testing and fluid sampling

RESULTS AND OBSERVATIONS

The amalgamation of UC 01 and UC 02 within the Site facilities has resulted in tangible operational and strategic advantages. The following are key findings and insights: -

Early Production Stage

The initial output from UC 02 wells, reintroduced into the Site facilities, exhibited oil production to test the wells' performance and potential. The produced fluid compatibility is confirmed through laboratory analysis and surface testing facilities.

Infrastructure Synergy

Leveraging shared pipelines, pumps, and production header led to a reduction in capital expenditures (CAPEX) by in comparison to independent UC infrastructure. This synergy facilitated:-

- Accelerated tie-in timelines (averaging **3.5 months** from well completion to full integration).
- Decreased operational redundancy and enhanced asset utilization.

Operational Coordination

The necessity of a RACI matrix among UC and Site teams is to foster improved coordination while minimizing handover delays. Significant operational milestones are expected to be achieved as scheduled, including:-

- Earlier commissioning.
- Clear identified battery limits for better communication.

Safety and Compliance

There are no recordable Health, Safety, and Environment (HSE) incidents during the integration process. This success can be attributed to:-

- Proactive identification of high-risk activities.

- Collaborative Hazard and Operability (HAZOP) reviews.
- Clearly defined emergency response protocols.

Strategic Value

This transformation synergy has showcased the practical potential of integrating unconventional resources into existing infrastructure, in line with COMPANY's goals for transformation by:-

- Accessing previously untapped reserves in UC 01 and UC 02.
- Enhancing the speed of market introduction for unconventional assets.
- Creating a scalable framework for future integrations of fields.

CONCLUSION

The integration of unconventional resources from UC 01 (Petronas) and UC 02 (Company) into Site site facilities marks a transformative milestone for Company. It exemplifies how strategic foresight, technical innovation, and cross-functional collaboration can unlock previously untapped value from existing infrastructure. By aligning operational capabilities with sustainability goals and accelerating production timelines, this initiative demonstrates a forward-thinking approach to resource optimization. The synergy between diverse teams and technologies not only enhances efficiency but also sets a precedent for future developments in unconventional resource management across COMPANY's portfolio.

Key takeaways include: -

- Seamless integration is achievable with the right studies, and communication frameworks.
- Early production from UC 02 validated the feasibility of unconventional-conventional synergies.
- The project has delivered tangible benefits in terms of cost, time, and safety.
- Lessons learned from this initiative are now being applied to future field developments across COMPANY.

This integration represents not just a technical milestone; it is a strategic catalyst that supports COMPANY's vision for a more adaptable, efficient, and sustainable energy future.

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